



Connecting fragmented enclaves through network? Managing industrial parks in the Jakarta-Bandung Urban Corridor

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ABSTRACT

This article seeks to explore the urban political and governance dynamics surrounding the planning and management of road infrastructure network between seven private industrial enclaves in Cikarang, the largest FDI-driven industrial parks agglomeration in the Jakarta-Bandung Urban Corridor (JBUC). We highlight the emergence of a new mode of governance that stresses the importance of network building among the industrial park developers and between the private and public sectors in addressing spatial fragmentation and infrastructure disconnectedness. Inter-actor networking strategies such as seeking of mutual interests and communication, cooperation and partnership have been initiated to reconcile the state-market divides and resolve the global-local political contestation. However, its success story in revitalizing the declining competitiveness of the industrial enclaves remains to be seen. This private-driven institutional arrangement finds difficulties in encouraging quick actions and building long-term commitment among its members mainly due to unbalanced power relation, fierce internal competition, and distrust among its members.

1. Introduction

Place-based governance has been criticized for its inadequacy in responding to the increased inter-territorial, inter-scalar, and inter-actor relations of our globalized world (Amin, 2004). New forms of governance are required to better accommodate the unbound, fluidity and interconnectedness of our current geographical sites, scales, and actors. Within the urban studies literature, network seems to emerge as a new typical urban governance form for managing suburban infrastructure fragmentation and helps reconciling its inter-scalar and inter-actor divides (Phelps, 2017; Young & Keil, 2010).

Considerable success stories of network governance can be found in the field of environmental management (Khan, 2013; Lo et al., 2018; Maes et al., 2018; Schulz, Lieberherr, & Zabel, 2018; Secco, Favero, Masiero, & Pattenella, 2017). Meanwhile, its exploration in urban regional context is rather complex in which embeddedness and local political competition contributes to influence the effectiveness and unique shapes of network governance (Blanco, 2015; Feiock, 2009). This article argues that, particularly in the case of suburban infrastructure along the Jakarta-Bandung Urban Corridor (JBUC), the transition toward network governance, without balanced power relation, cooperative reputation, and sense of moral obligation can be far from an

easy task.

This article assesses the effectiveness of network as an alternative governance form in coordinating urban transport infrastructure in Cikarang, the largest and oldest concentration of private industrial enclaves in the JBUC, 60 km east of Jakarta and 100 km northwest of Bandung. It addresses several interrelated questions: to what extent has network governance been initiated in Cikarang? Has it been effective in facilitating more sustainable urban enclave transformation? What factors have played a role in fostering or hindering the effectiveness of this network governance?

Addressing these questions, semi-structured interviews were conducted with those directly involved in the spatial and governance transformation in the Cikarang industrial corridor since the 1980s. In total, 30 individual and group interviews (four of which were follow up interviews), each lasting for about 1 h, were conducted between 2012 and 2016. There were 27 interviewees consisted of nine local government officials, five provincial government officials, three central government officials, and ten private industrial town planners, developers and managers. The interviews mainly prompted questions concerning the respondents' background, views, and own experiences in the building of network governance in Cikarang. Interview analysis was conducted to explore the dynamics of competition, conflicts,

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confrontations, and cooperation among the different participating groups in transport infrastructure management in Cikarang. As a triangulation, we compared different sides of the story from minutes of meetings, policy and legal documents, newspaper and online news network. At the same time, field observations were carried out to describe the impact of governance on the progress of transport plan implementation in the industrial park agglomeration.

The presentation of this article is structured as follow. First, we provide a critical review of urban enclave governance and network governance theory as a framework of analysis in this article. Next, we explain Indonesia's historical experience in developing industrial land and policy. It includes the role of the policy in producing urban enclave and infrastructure fragmentation in Cikarang Industrial Park Agglomeration. The next section explains the potential emergence of network as a new mode of governance in addressing spatial divides and infrastructure fragmentation. In the analysis, we highlight the ineffectiveness of governing through network and explain its hindering factors. The last part concludes the discussion.

2. A critical review on urban enclave governance

While social and spatial features of urban or industrial enclave are widely recognized, their organizational, institutional, and governance issues are seen to be the most contested area of research (Phelps, Atienza, & Arias, 2015; Roitman, Webster, & Landman, 2010). A debate on these issues includes the selection of appropriate spatial scales at which place-based governance can effectively play (Sood, 2015). An early viewpoint has focused on the *micro*-structure and processes of governance within the project site or at the “enclave” scale. At this level, privatization modes of planning and urban management taking the forms of residential “club”, homeowners' association, corporate city, and industrial township are often highlighted (Chen & Webster, 2005; Denis, Mukhopadhyay, & Zerah, 2012; Murray, 2015; Webster & Glasze, 2004). A later viewpoint has underlined the importance of *macro*-institutional factors, which primarily consist of government regulatory and policy frameworks at the state scale and national cultures, in facilitating the construction of urban enclave (Tedong, Grant, & Wan Abd Aziz, 2015; Zhou, 2014).

Adopting both extreme views instantly to the current Indonesian context can be problematic. At the macro scale, since the implementation of a radical decentralization policy in the early 2000s, the national hierarchical state's intervention in urban planning, development and management sectors have relatively become more minimized than that of during Suharto's authoritarian regime. At the micro (or the industrial park project) scale, the growth of several private industrial parks in the JBUC, Indonesia's largest urban corridor, has exceeded their internal institutional and resource capacity and thus generated externalities to their surrounding areas. As a result, the private industrial developers recently face difficulties to continue providing their tenants with premium land resource, infrastructure and facilities in an efficient and competitive way.

Hence, as an alternative, by using the network governance perspective, this article emphasizes the role of governance processes that cut across the macro-micro and state-private divisions and have balanced implications for the *meso* scale, or the urban and regional scale. Such an alternative perspective is expected to help fill the gaps, recover the “institutional missing”, and mediate the relationship between the micro (enclave) geographical scale and the macro (nation state) geographical scale on the issues such as urban infrastructure provision and spatial planning and management.

3. Network governance

Network as a form of governing structure and mechanism was first developed in the organization and management science literature. Network can broadly be defined as a horizontal social relationship

(Powell, 2003). From a transaction cost economic perspective, network can be seen as a response to market and internal organizational failures in coordinating and safeguarding exchanges of products and services within and between autonomous firms (Williamson, 1973). Therefore, in analyzing the emergence of network governance, we focus on several basic criteria, which at the same time can be contrasted with market and hierarchy: membership, objective, coordinating mechanism, and exchange.

The *first* criterion is about network membership. As Jones, Hesterly, and Borgatti (1997) highlight, network consists of legally “autonomous” organizations so that each organization is relatively independent in making their own decision. They maintain that network participants are selective, thereby normally neither restricting nor encompassing an entire population. Theoretically, a network comprises a minimum of a group of three organizations (Provan & Kenis, 2008). The number of participants in a given network is considered important in increasing overall coordination costs (Kim, 2006). In the case of public sector, Kim (2006) suggested that, in order to be effective, an optimal size for a “small” network should not more than six to eight organizations.

The *second* criterion is network objective. Provan and Kenis (2008) maintain that a network is formed to attain not just each participant's goals but also collective goals. Compared with hierarchy, participants within a given network have different goals and orientations. However, regardless the differences, they are able to build common interests that encourage them to distribute their resources evenly, a result that cannot be found in market.

The *third* criterion is coordinating and controlling mechanism. In contrast with the market and hierarchical governance structures, network governance emphasizes an inter-organizational coordinating mechanism typified more significantly by informal and horizontal social relationships than formal contracts and hierarchical bureaucratic structures (Jones et al., 1997; Powell, 1990). Network governance is a wider approach than public-private partnership. In fact, as an informal arrangement, network is able to overcome the limitations of the formal arrangement in the public-private partnership in managing inter-actor relationship and resolving conflict. In controlling the behavior of parties and resolving conflict, network governance is more dependent on normative and socio-cultural mechanisms such as collective sanctions and reputations, than on authority and legal recourse (Jones et al., 1997; Powell, 1990).

Finally, in contrast with market and hierarchy, communication and exchange within a network are neither random nor uniform but are rather patterned thus reflecting a division of labor (Jones et al., 1997; Powell, 1990). Network governance is typified by reciprocal patterns of communication and exchange. Powell (1990) underlines that the concept of reciprocity is central to network governance. He refers reciprocity to a relationship created based on indebtedness, a measure of imbalance that compels parties to recur meeting and exchange, and a sense of obligation through which participants remain connected to one another.

Network governance is deemed to mark a fundamental shift in government, where the governance process is pluralistic due to the increased involvement of non-governmental actors (Ansell & Gash, 2008). Network is also expected to be an effective governing form in dealing with complex issues associated with fragmented, uncertain and competitive environments (Jones et al., 1997; Koppenjan & Klijn, 2004). Nevertheless, our case study of road infrastructure in Cikarang shows that stimulating a network governance to grow is not an easy task. In line with Blanco (2015) and Wachhaus (2012), in the following case study and analysis we assess the effectiveness of network governance and explore the local and contextual factors that enable or hinder its development.

4. Setting the context: industrial land development policy in Indonesia

Indonesia's experience in developing industrial area started with an industrial zone, which refers to unplanned concentration of industrial activities in designated locations. Industrial zone in Indonesia is commonly occupied by local and labor-intensive industries while foreign and hi-tech industries tend to operate in planned industrial park (Hudalah, Viantari, Firman, & Woltjer, 2013). The industrial park is built as a response to a growing demand for industrial land, pollution caused by industrial activities, lack of infrastructure, and the encroachment of residential areas near the industrial locations. In the early 1970s, industrial parks in Indonesia were mainly built and operated by state-owned enterprises as an attempt to attract the local and foreign capital investment. In 1973, Jakarta Industrial Park Pulo Gadung (JIEP) as the first industrial park was built. In the following year, Surabaya Industrial Park Rungkut (SIER) was also built. During the early decades of industrial park development, state-owned enterprises built approximately five industrial parks i.e. Cilacap Industrial Park (1974), Medan Industrial Park (1975), Makassar Industrial Park (1978), Cirebon Industrial Park (1984), and Lampung Industrial Park (1986).

In a later period, market-oriented and outward-looking policies issued during the New Order regime led by former President Soeharto had succeeded to accelerate economic growth by boosting FDI in the manufacturing sector. Those policies stemmed from the government's vision to reduce economic dependency on oil and gas revenues yet to attract domestic and foreign investors. In view of that vision, the global-scaled export-oriented industries relocation from Asian industrialized countries, especially Taiwan, Japan, Hong Kong, South Korea, and Singapore, had been captured by Indonesia's government (Wie, 1991).

The Soeharto's macro policies attracted the private sector to participate in industrial park development. Along with a growing number of proposals from domestic and foreign private investors, through Presidential Decree 53 (1989) about Industrial Park, the government allowed industrial park to be managed by both domestic and foreign private parties, with or without the participation of state-owned enterprise. This presidential decree has simplified previous regulations thus creating favorable conditions for privatization and foreign investments. This policy aimed to encourage the planned and advanced development of industrial activities by occupying the designated areas. With this presidential decree, the government allowed the private companies to self-regulate and enclose public space through privatized development. Furthermore, private developer began to get involved and dominating the development of suburban industrial park (Hudalah et al., 2013).

Other presidential decrees were established to set rules for industrial park development. Presidential Decree 33 (1990) about Land Use for Industrial Development Area and Presidential Decree 98 (1993) about Amendment of Presidential Decree 53 (1989) about Industrial Park were mainly focused on land use for industrial parks. Private developers had to build in selected locations intended for industrial activity in accordance with district or municipal spatial plan prepared by local government. These regulations can be seen as the government's measures to control the development of industrial parks by private developers. In order to reinforce the regulation on industrial park development, Presidential Decree 41 (1996) about Industrial Park emphasized that industrial park as a concentration of industrial activities has to be equipped with infrastructures and supporting facilities developed and managed by licensed industrial park companies.

The enactment of Law 22 of 1999 on Regional Administrations and Law 25 of 1999 on Inter-Government Financial Balance marked the beginning of *otonomi daerah*, or the Indonesian decentralization policy, led to Indonesia's radical shift from a centralized, authoritarian regime toward a highly decentralized administrative system. This decentralization policy was considered very ambitious as it entailed the devolution of a wide range of political powers in most policy areas,

including industrial land use planning and local road infrastructure development, occurred even to the lowest administration level or local government. Furthermore, the independent role of the local council-elect or *Dewan Perwakilan Rakyat Daerah* (DPRD) was strengthened in making laws and holding accountability of the head of the local government. Law 25 of 1999 also gave the local government an authority to exercise power over local revenues through taxes, charges, and fees from local businesses and citizens (Kimura, 2010). As the result, the local policies could even be executed without any requirement for close consultation with the central and provincial government administrations. Many autonomous regions including the Bekasi District have designed their own local regulations and policies concerning investment, land, and property development. Such policies may potentially complicate procedures and, thus, creating uncertainty and increasing transaction costs for industrial investment and production in industrial parks (Hudalah, 2017).

In the past two decades, there has been rebalancing phases in the decentralization policy. Law 22 of 1999 was amended twice, first was by Law 32 of 2014, then the Law 23 of 2014. The later retracts authorities on coastal and forestry management, which previously devolved to district/municipal government, to the provincial government. Law 25 of 1999 was also amended by Law 33 of 2014, which gives an increased share of revenues to the regions allocated for basic education. Clearly, regardless these amendments, the overall policy does not revoke the remarkable authorities given to district/municipal government in the key areas of local land use planning, property development, and infrastructure management.

5. Case study overview: governing the roads through network

Located in the central part of Bekasi District, Cikarang is the largest concentration of manufacturing activities in JBUC., which captured almost 60% of the country's economic activities (Hudalah & Firman, 2012). Since 1989, seven private industrial parks and towns totaling almost 16,000 ha have been established in the industrial park agglomeration (see Fig. 1). Jababeka (5600 ha) and Delta Silicon/Lippo Cikarang (3500 ha) are owned by Indonesia's large developers. Jababeka is the largest and only private industrial park and town in Cikarang that show a high level of diversity by attracting small and medium-sized companies from more than 30 different countries. Meanwhile, five other industrial parks are managed by joint venture companies combining domestic and foreign investors. Their tenants tend to be segregated by the nationalities of their principal foreign investors. BFIE/Bekasi Fajar (2100 ha), MM2100 (1200 ha), EJIP (320 ha), and GIIC/Deltamas (3000 ha) are dominated by groups of select large-scale multinational companies from Japan. Likewise, the multinational companies from Korea tend to concentrate in Hyundai Inti Development/BIIE (200 ha).

The management of road infrastructure in Cikarang is complex as it involves many parties from both government and the private sector. Under the decentralization policy, the authority to build and manage road networks in Cikarang is distributed among the central, provincial and local governments. At the national level, Ministry of Public Works and Public Housing is the technical ministry in charge of the development and maintenance of the primary arterial road that connects major cities in the Java North Coast (Pantura). PT. Jasa Marga is the state-owned enterprise that owns many toll road concessions including the Jakarta-Cikampek Toll Road, which divides Cikarang into the northern and southern industrial park agglomerations. West Java Provincial Government oversees the development and maintenance of a number of primary collector roads, which connect regional cities in the northern, middle, and southern part of the Island. At the local level, Bekasi District has the authority and responsibility to build most road networks in Cikarang, consisting local and secondary roads that connect all towns in their jurisdiction. Meanwhile, all roads located in the industrial parks are managed by their respected private developers.

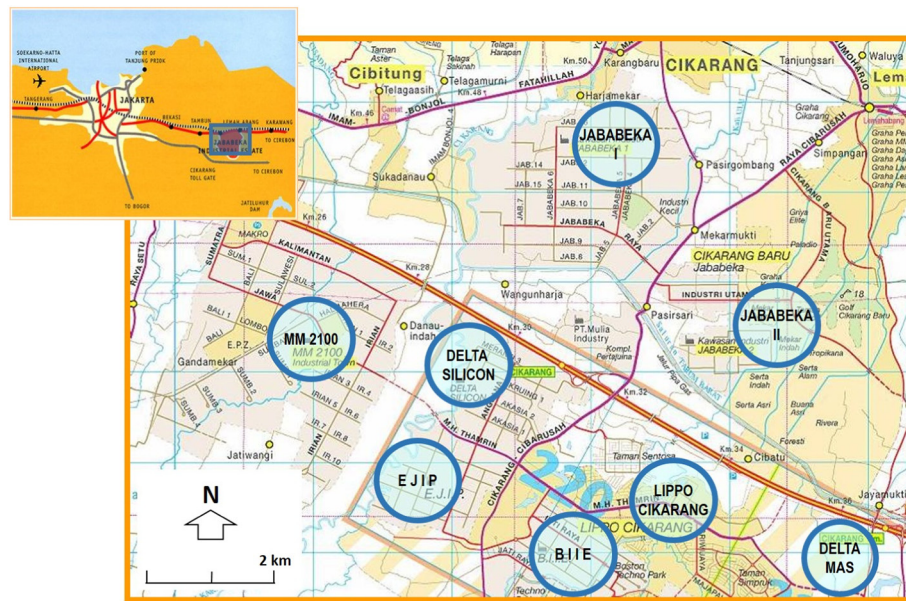


Fig. 1. The seven industrial parks in Cikarang.

Source: Jababeka, 2007

The decentralization policy has forced the central government to withdraw their involvement in local development affairs and left Bekasi District with limited fiscal capacity. While their jurisdiction had become the prime location for manufacturing industries, their headquarters and main offices were still located in Jakarta. Consequently, Jakarta holds the largest shares of the income and company taxes. Meanwhile, the district government can only extract much smaller shares, particularly from land and building taxes. As the result, despite the decentralized authority, the local government lacks the institutional and financial resources, which inhibited them from provisioning good infrastructure and facilities to support the operation of the industrial parks (Hudalah, 2017). In contrast, the private sector has no authority but stakes and capacity to develop infrastructure within and surrounding their project sites. This condition potentially creates an interdependent relationship between the government as the authorized institution and the private sector as the capital holder, a common situation that, according to Jones et al. (1997), potentially encourages a network governance to develop.

As the first attempt to build a coordinating institution, in 1993, the central government through the Industrial Parks Coordination Team initiated a handshake agreement with the provincial and local governments and industrial park developers in Cikarang in pursuit of resolving the local infrastructure fragmentation. This public-private partnership agreement required industrial park companies to build connecting roads with their neighboring project sites and local communities. However, in the implementation, it seemed that most companies undermined this agreement.

From the beginning, it is well noted that the industrial park developers in Cikarang had given a high priority on the provision of excellent infrastructure networks within their own project sites, causing each industrial park project to become isolated from one another (Hudalah, 2017). Each industrial park focused on the management of internal road networks within their site and on the improvement of their access to the Jakarta-Cikampek toll road as the main access to the International Port of Tanjung Priok. Consequently, many industrial park projects became inadequately connected with its neighboring projects.

As the turning point, in 2006 Jababeka, the largest and oldest industrial park developer in Cikarang, reinvented six other industrial park managers to initiate the establishment of an association called *Zona Internasional* (ZONI) or International Zone (see Fig. 2). The developer's

initiation was formalized in the form of a handshake agreement (MoU) backed-up by the central government (the Ministry of Public Works), West Java Provincial Government, Bekasi District Government, and the state-owned highway corporation (PT Jasa Marga). This network organization had the vision to create a special zone that can offer a good environment for sustainable investment and development acceleration in Cikarang. It had the purpose to develop and implement policies; manage development programs and public assets in order to revitalize the competitiveness of the industrial park agglomeration.

As part of the agreement, a non-binding action plan was made to improve the road infrastructure condition in order to integrate the operation of the seven industrial parks. The participating actors, both from government and private sectors, played complementing roles in the transport infrastructure development (see Fig. 3). The central government focused on building of new toll roads, flyovers, and bridges as part of the primary arterial road networks in Java's North Coast. They were responsible to conduct a feasibility study and to build seven new infrastructures, including Cibitung (Cikarang)-Cilincing toll road, Cimanggis-Cibitung toll road, Tanjung Priok Access Toll Road, Cikarang-Cibarusah fly over, and bridges at MH Thamrin street. They also improved Bekasi-Cikarang national road Cibarusah Raya (Lemahabang-Mekarmukti) Street. The building of five new toll road lanes on the existing Jakarta-Cikampek Toll Road became the Indonesia Highway Corporation's (PT. Jasa Marga). The provincial government was responsible to build the main roads while the local government built the local roads. Meanwhile, the industrial park developers carried out the construction and maintenance of two road networks that connect all southern industrial parks in Cikarang (MM2100-Lippo Cikarang and Lippo Cikarang-Delta Mas) (Jababeka, 2007).

6. Analyzing the (in)effectiveness of the network governance

In its initial version, the action plan demanded the road infrastructure projects to be completed between 2007 and 2011. However, during its implementation, most projects were delayed. By 2009, PT Jasa Marga, the state-owned highway corporation, and provincial government have fulfilled most of their responsibilities. Meanwhile, several identified central government's projects were not yet completed. The development of new toll roads connecting Cibitung (Cikarang)-Cilincing and Cimanggis-Cibitung were delayed due to difficulties in

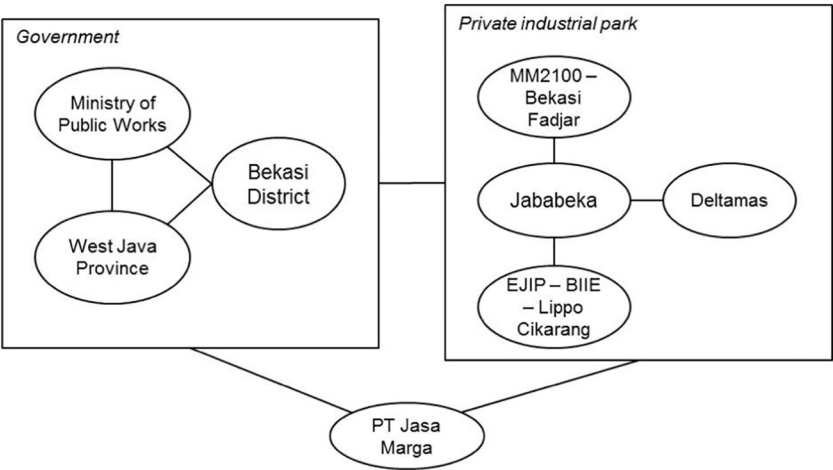


Fig. 2. Interactor relationship in ZONI.
Source: own analysis

land acquisition. Suspended implementation was also experienced by the development projects carried out through public-private partnership (PPP), which was supposed to be finished by 2009. These PPP projects include the development of new interchanges between Cibitung and East Cikarang (partnership between the industrial park managers and the central government), the development of Cikarang-Cibarusah (MH. Thamrin) underpass (partnership between the industrial park managers and the provincial government), and the development of new roads connecting the industrial parks and nearby primary arterial roads (partnership between the industrial park managers and the local government). In response to the difficulties in the implementation, especially the connecting road projects, plan revisions were made in 2010 and 2012. The revisions attempted to accommodate the private developers' proposals to include new items and extend the deadline of projects' accomplishment. Despite the extended deadline,

Table 1
The implementation progress of ZONI Plan (2016).
Source: updated from Hudalah and Firman (2017)

Responsible actor(s)	Finished	In progress	No progress
Central government	6	2	
PT Jasa Marga	4		
West Java Provincial Government	3		
Bekasi District Government	1	1	
Industrial-park managers	1		1
Public-private partnership			3
Total	15	3	4

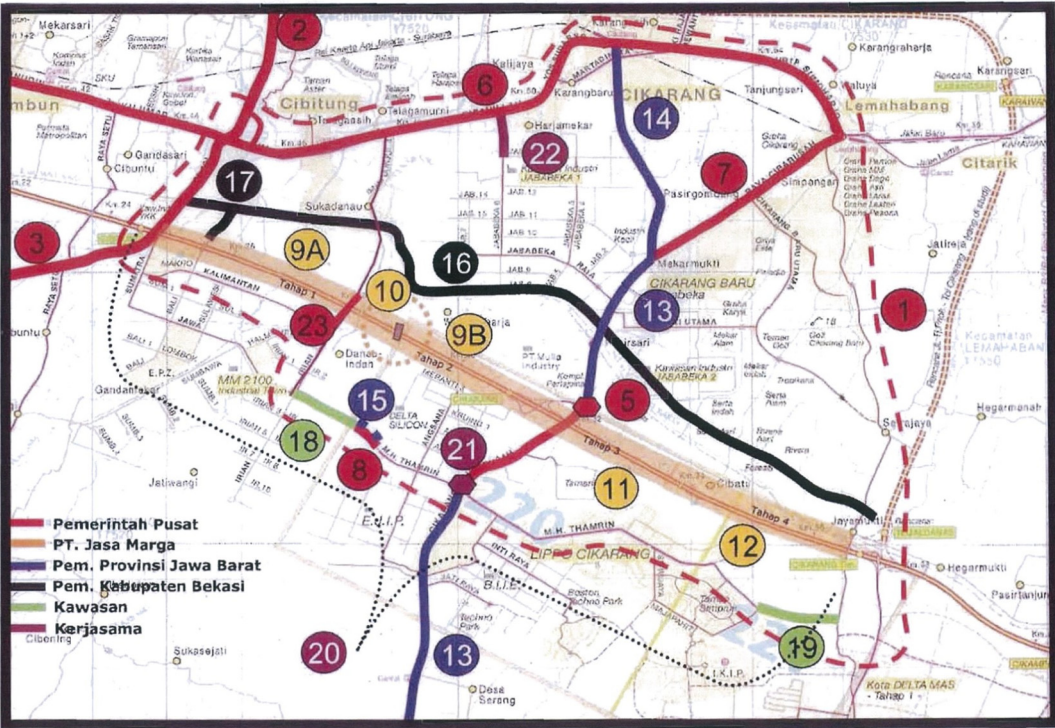


Fig. 3. International Zone (ZONI) Plan.
Source: Jababeka (2007)

by 2016, seven of all 22 projects had not been implemented or completed (see Table 1).

Based on the agreement, the main responsibility of each industrial park developer was to build connecting roads between their own project site and their neighboring ones. The developers were given wider responsibility to build connecting roads located outside their project sites in order to reduce the ineffectiveness of the unbundling infrastructure previously developed by each of them. In practice, several industrial park developers did not cooperate well in carrying out this responsibility. To illustrate, there were five of 22 projects conducted through public-private cooperation and private-private cooperation. However, only one project was ever completed and, even so, it failed to meet the agreed extended deadline.

Two notable case studies of connecting road constructions can best illustrate the complexity of inter-actor relations leading to the dynamic and delayed implementation of the action plan: MM2100-EJIP and Lippo Cikarang-Deltamas.

6.1. Case study analysis 1: MM2100-EJIP connecting road construction

The first case included a conflict occurring between MM2100 and EJIP, two neighboring industrial parks separated by Cikarang River. Aiming to connect the two neighboring industrial parks, in 2010, the central government completed the construction of two units of 60 m bridge crossing over the river (Fig. 4). The government expected that the bridge construction would be followed by the construction of a connecting road from each side of the industrial park sites. Realizing that the bridge would help shorten their distance to Jakarta International Seaport, EJIP quickly completed the road construction from their part. Meanwhile, MM2100 is located closer to Jakarta compared with its competing neighbor, EJIP. From the business perspective of MM2100, the road construction, therefore, would only divert the toll road traffic between EJIP and Jakarta:

“... most companies seemed to undermine this agreement. They argued that connecting roads would generate through traffic on their local roads thus causing congestion and increasing the road maintenance costs. Such an externality would only benefit their neighboring competitors who wish to get a free and shorter access to nearest interchanges that connect their project locations with

Jakarta”

(Interview with MM2100's manager, 27 June 2016)

Largely due to this business competition, MM2100 postponed the road construction for more than a decade. By 2016, MM2100 started the construction but its progress was very slow. They maintained that the construction was also hindered by a high cost of land acquisition.

6.2. Case study analysis 2: Lippo Cikarang-Deltamas connecting road construction

The second case study is concerned with the construction of the road connecting Lippo Cikarang and Deltamas. The road construction was crucial to give the district's new capital a direct access to other areas under their jurisdiction. As part of the handshake agreement with the district government, for Lippo Cikarang, the road construction would serve as the ticket to start building Pondok Gede Interchange, which would provide their project site and the seat of the district government a direct access to the inter-city toll road. However, Lippo Cikarang deliberately deferred the road construction. The developer argued that there were no economic needs for its development (Interview with urban designer of Lippo Cikarang, 3 April 2013). In fact, they claimed that such initiative would only harm their business interests because they had to pay for maintenance costs of the road that generated no income. Furthermore, the opening of the connecting road would just benefit their closest competitor by shortening the travel distance of Deltamas's tenants to the seaport. The postponement has created conflicts between Lippo Cikarang and the local government. As the result, the government eventually revoked their agreement and relocated the planned location of the interchange to Deltamas industrial park, namely Cikarang Timur Interchange (37th kilometer of the Jakarta-Cikampek Toll Road) (Interview with a former head of Bekasi District Regional Development Planning Agency, 14 June 2016).

The government's need for a new interchange was solved but the conflict between Lippo Cikarang and Deltamas concerning the opening of inter-industrial-park connecting road continued to persist. During its critical period, the local government was forced to send an attorney to settle disputes with the industrial-park manager (Hudalah & Firman, 2017). As the result of the negotiation, in 2012 Lippo Cikarang requested the government to allow them to build another toll road



Fig. 4. Bridge connecting MM2100 and EJIP.
Source: own observation (2016)



Fig. 5. Gated road connecting Lippo Cikarang and Deltamas.
Source: own observation (2016)

interchange within their project site, which was later named Cibatu Interchange, in exchange for the connecting road construction (Fig. 5) and the relocated interchange. Previously, Cibatu Interchange was not listed in the ZONI Plan but, later, the developer was able to coerce the government to agree on drawing the new item in the revised plan. In 2014, following a lengthy decision-making process, the local government finally granted the permission for the opening of the new interchange right after Lippo Cikarang completed the construction of the inter-connecting road to Deltamas. The new interchange is located on the 34th kilometer of Jakarta-Cikampek Toll Road, which is only 3 km apart from Cikarang Timur Interchange located in Deltamas.

7. Discussion: why network fails

The building of network governance in Cikarang industrial agglomeration faced many challenges. The strategies in seeking mutual interests, reciprocal benefits, common problems and initiating communication, cooperation and partnerships find difficulties to be implemented. Several conditions and factors have contributed in constraining the governance through network as discussed below.

7.1. Unbalanced power relation

In line with Powell (2003), as a form of network, the working relationship in Cikarang industrial agglomeration was built horizontally and not rigidly. The networking actors included those had interests and influence in the road infrastructure network in Cikarang. They also shown different resources capacities, based on which, as scholars have emphasized, an effective network might be built (Kim, 2006; Koppenjan & Klijn, 2004; O'Toole Jr, 1997; Powell, 2003). The governments became the only institution that is eligible in formulating policies and regulations related to road infrastructure. Meanwhile, the private sector has the capacity to bring jobs and investment into the area.

Nevertheless, partly due to inequality of financial and institutional capacities owned by different parties, this differentiation did not result in an effective network as the literature have suggested. As the result of partial fiscal decentralization, the district government was given a wide range of spending authority but has limited power to generate their own income, resulting in a poor performance of the local public services

to fulfill the industrial park needs. Furthermore, the central government treated industrial park as a special zone whose infrastructure master-planning and management are given to licensed company. Given its weak financial capacity and authority, the local government was unable to play properly as the coordinator of infrastructure development in their own area. Meanwhile, due to their very high financial capacity and excessive control on the industrial land, the industrial park companies had a great power to influence and direct the decision-making and the development of Cikarang industrial agglomeration. Particularly, Jababeka, as the oldest and largest private industrial park manager in Cikarang, had a close relation with the central, provincial and local governmental actors, making them possible to play as the initiator and leading actor in the formation of ZONI.

7.2. Internal competition

The consensus to build ZONI was a process whereby all actors participating in the decision-making contributed to determine road infrastructure programs to overcome inefficiency caused by physical fragmentation in Cikarang. It resulted in an agreement that accommodated the interests of involved actors. *Initially*, the consensus was able to build a commitment among the governments and the industrial park managers to address collaboratively the congestion problems. However, previous interactions spearheaded a closed and suspicious attitude among the private developers. The experiences do not show a reputation of long-term cooperative relationship. Meanwhile, recurring mutual exchanges with long-term perspective is considered important to create the reciprocal pattern of relationship, the central concept in network governance theory (Jones et al., 1997). Historically, the private developers in Cikarang were far from partners and, instead, competed to attract and secure the same types of industrial tenants, who constituted their shared market and consumers. Although ZONI was initiated under a common interest of improving the competitiveness of the industrial agglomeration as a national growth center, the business interests among the private developers remained and could not be ignored. As can be seen in the case study of conflict between MM2100 and EJIP, several private developers still worried about the externality costs of building inter-connecting roads. Consequently, the relationships among the actors were no longer built on behalf of long-term

common interests but on individual's own short-term ambitions.

7.3. Opportunism and distrust

The interaction between the government and private developers had started since the development of industrial park through the division of tasks in infrastructure provisioning. Nevertheless, it was not an amicable interaction. When it comes to industrial infrastructure management, the local government's reputation was poorly rated by the private developers. As shown in the case study section, the government was considered to have a lack of capacity in dealing with the global constellation of industrial parks and provide the needs of investors as the tenants of the industrial parks. The private developers were also worried about the inability of local government to ensure the business convenience and efficiency and security of tenure in the industrial parks following the opening of the connecting roads. These worries and fears imply an absence of trust among the developers on the local government. Trust or sense of ethical obligations plays an important role in strengthening the collaboration among networked actors (Provan & Kenis, 2008). It resolves complex disputes far more quickly and economically than coercion, authority, or bargaining (Powell, 1990). The absence of trust between the private developers and the government had become a major constraint in the implementation of ZONI to improve the infrastructure within Cikarang Industrial parks and the surrounding area.

From the government's perspective, opportunistic behaviors of the private developers were deemed to contribute in generating distrust among the network's members. The initial analytical section illustrates that while the government had completed almost all infrastructure projects under their responsibilities as agreed in the plan, most industrial park managers could not show their significant commitment in carrying out their parts in time. The private sector would only construct and were willing to open the connecting roads' gates after their short-term business interests were accommodated in the MoU. An example of developers' opportunism can be seen in the MoU's implementation. The case study of the connecting road between Lippo Cikarang and Deltamas suggests that the private sector often incidentally proposed adding new items or conditions (such as building of new interchange) to better meet their own latest interests. Such inconsistent decision has caused conflicts and distrust, following which other actors would not want to continue the cooperation or undermined their commitment and responsibilities as members of the network.

8. Conclusion

In the case of Cikarang industrial park, *Zona Internasional* (ZONI) or the International Economic Zone can be seen as an example of how network governance was initiated to deal with the complexity of globally-linked suburban infrastructure management. As a growing network during the 2000s, ZONI's members consisted of few selected autonomous organizations from the private sector and government agencies. The declining competitiveness of Cikarang as the JBUC's manufacturing powerhouse had forced some members to sit together to build the common objective of resolving the spatial fragmentation and improving the inter-enclave connectivity in order to create a good environment for sustainable investment and development acceleration. The coordination in the network was conducted through informal communication among the participating actors. Person-to-person communication and thematic forums were designed as the key vehicles for information exchange, collaboration initiation, and social control. The decision-making process in the network governance was initially encouraged to be consensus-oriented whose outputs included non-legally binding action plans.

However, the article shows that the building of the network governance is not an easy task. It reveals that a rural local government under immature decentralization lacks the institutional and financial

capacity to build the road infrastructure that could meet the international standard of manufacturing production and distribution. Meanwhile, the private industrial park managers have a much higher financial capability and closer relationships with higher tiers of governmental actors. Such an inequality of resource capacity created unbalanced power relation, based on which horizontal communication could not be used as an effective tool to maintain the sustainability of the network.

In addition, short-term business interests and competitive relationships among the neighboring industrial parks allowed opportunistic behaviors to dominate their course of action. The private sector had foiled the agreement as stated in the plan but did not show serious commitment during its implementation. Under the pressure of competition among themselves, the developers might hesitate to cooperate and, thus, undermine the agreed collective goals and the division of tasks outlined in the plans. Instead, they focused on the fulfillment of egocentric goals by building exclusive infrastructure mostly located within their own project sites. This egocentric behavior created conflict and distrust among the participating actors, which in turn undermined the norm of reciprocity and, thus, challenged the effectiveness of the network governance. If any, trust was not evenly distributed among all participants, which hindered them to be able to move as a collective entity.

In order to improve the effectiveness of network governance in the context of fragmented, multiscaled, and globalized suburb, several policy recommendations can be put forward. First, in the absence of trust and norm of reciprocity, a leader needs to be appointed or promoted to provide an alternative basis for collaboration. As Provan and Kenis (2008) suggest, such leader has the special tasks of absorbing transactional costs, resolving internal conflict, and strengthening members' commitment. In fulfilling the tasks, as the second policy recommendation, the network leader should not rely on directing and monitoring mechanisms but strengthen coordination functions and help identify the instruments that can motivate the network to address the distrust issue (Cavelty & Suter, 2009). Due to conflict of interests and egocentric behavior, we should not give this special role to the private sector. Instead, local or regional government is the most legitimate actor to take on such a role. To this end, our last policy recommendation is to build the institutional and financial capacity of the subnational government so that it can better meet the burden of authorities caused by the decentralization policy. With this newly built capacity, the government will be able to improve their bargaining position among other established members of the network governance and, thus, be able to play a central role in managing the flow of key resources as the basis for maintaining the sustainability of the network.

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